

A Appendix

A.1 Dataset

Moral Foundation	N	Rule-of-Thumb	Action
authority-subversion	89	2.98 ± 0.08	2.99 ± 0.08
care-harm	93	3.22 ± 0.09	3.16 ± 0.09
fairness-cheating	89	3.09 ± 0.10	3.12 ± 0.08
loyalty-betrayal	89	3.01 ± 0.08	3.06 ± 0.08
sanctity-degradation	90	3.11 ± 0.08	3.13 ± 0.09

Table A1: Mean Rule-of-Thumb (RoT) and Action agreement scores ($\pm$ SE) for each Moral Foundation. The RoT agreement reflects how much people agree with the general principle stated in the rule, while the Action agreement reflects how much people agree with the judgment about the specific behavior described in that rule. Scores use a 0-4 scale (0 = rare agreement, 4 = near-universal agreement), indicating the estimated proportion of people who would express that agreement.

A.2 Survey

A.2.1 Instructions for Human Participants

In this study, you will read a brief statement describing a person’s behavior. Then, you will answer a few questions about how this person might feel and how others might feel or react after seeing the behavior.

You will only see the behavior description once at the beginning of each set of questions. For each behavior, you will answer several questions one at a time across separate pages.

If you forget the behavior, you can click the “**Show Reminder**” button to see it again on any page.

Here are some things to keep in mind while judging the scenarios:

- The agents performing the actions are U.S. adults.
- Each scenario is about different individuals, even though they might share the same name.
- The names are randomly generated to keep the actors anonymous.
- If anything seems unclear, please use your best judgment or follow your intuition.

When you are ready, please click the “»” button below to begin the study.

A.2.2 Compensation

Participants received monetary compensation of \$12/*hr* for participating in the study.

A.2.3 Questions

Actor Emotion

How would {the actor} feel about their action? If they feel multiple emotions, select all that apply:

- Shame (1)
- Guilt (2)
- contempt (3)
- Anger (4)
- Disgust (5)
- Embarrassment (6)
- Pride (7)

- Compassion (8)

Follow-up: For each selected emotion, please rate the intensity of each emotion {the actor} would feel.

1 = A little, 2 = Moderate, 3 = A lot, 4 = Extremely

Observers' Emotions

Answer for each type of observer:

Close friends or relatives

Suppose {close friends or relatives} observed or were aware of {the actor}'s behavior. How would they feel about the action?

Response choices: same as Actor Emotion

How strongly would close friends or relatives feel each selected emotion?

1 = A little, 2 = Moderate, 3 = A lot, 4 = Extremely

Acquaintances

Suppose {acquaintances} observed or were aware of {the actor}'s behavior. How would they feel about the action?

Response choices: same as Actor Emotion

How strongly would acquaintances feel each selected emotion?

1 = A little, 2 = Moderate, 3 = A lot, 4 = Extremely

Strangers

Suppose {strangers} observed or were aware of {the actor}'s behavior. How would they feel about the action?

Response choices: same as Actor Emotion

How strongly would strangers feel each selected emotion?

1 = A little, 2 = Moderate, 3 = A lot, 4 = Extremely

Observers' Reactions

Answer for each type of observer:

Close friends or relatives

How would the actor's close friends or relatives react toward the actor's action?

1. Do nothing
2. Stay away or keep distance from the actor
3. Gossip about / discuss with / complain to others informally
4. Verbal reprimand / verbal confrontation
5. Physical confrontation
6. Inform the authorities, such as the police
7. Support the actor
8. Celebrate / praise

How should the actor's close friends react toward the actor's action?

1. Do nothing

Model	Release	Cutoff	Context
<i>Closed-source models</i>			
GPT-5.2	Dec'25	Aug'25	400K
Claude-4.5-Opus	Nov'25	Aug'25	200K
Gemini-3-Pro	Nov'25	Jan'25	1M
<i>Open-source models</i>			
GPT-OSS-20B	Aug'25	Jun'24	128K
LLaMA-4-Scout-17B	Apr'25	Aug'24	1M
Gemma-3-12B	Mar'25	Aug'24	128K

Table A2: Release dates, training cutoff dates, and context window sizes for the large language models evaluated in this study. Open-source model parameter counts are included in the model names.

2. Stay away or keep distance from the actor
3. Gossip about / discuss with / complain to others informally
4. Verbal reprimand / verbal confrontation
5. Physical confrontation
6. Inform the authorities, such as the police
7. Support the actor
8. Celebrate / praise

Acquaintances

How would the actor's acquaintances react toward the actor's action?

How should the actor's acquaintances react toward the actor's action?

Strangers

How would strangers react toward the actor's action?

How should strangers react toward the actor's action?

Norm Judgments

Some people think the actor's action is appropriate, and others think it is inappropriate. To what extent do you think the actor's action is inappropriate?

1. Not at all inappropriate
2. Somewhat inappropriate
3. Inappropriate
4. Very inappropriate
5. Extremely inappropriate

Some people think the actor's action is right, and others think it is wrong. To what extent do you think the actor's action is wrong?

1. Not at all wrong
2. Somewhat wrong
3. Wrong
4. Very wrong
5. Extremely wrong

A.3 Prompt

For the metanorm prediction task, we provided all models with the same set of 16 questions asked in human survey in a fixed schema and a shared instruction template. Unlike the human survey experiment, we let LLMs report other emotions not included in the predefined list. The general prompt was the same across all six LLMs, with minor model-specific formatting tweaks. The prompt used is provided below.

You are participating in an academic research study on moral judgments. The scenario may involve sensitive topics, but they are purely hypothetical and for research purposes only. You are given a statement about a behavior:

{stimuli}

The actor is: {actor}.

Your task is to answer a fixed set of 16 questions about this behavior. Each question has clearly defined answer choices.

Instructions:

- For most questions, use **ONLY** the provided answer choices.
- Do not rephrase or alter the provided options.
- For emotions (Q1, Q3, Q7, Q11):
 - If “No emotion” is selected, it must be the **ONLY** choice, and the corresponding intensity question (Q2, Q4, Q8, Q12) must be null/NaN.
 - If the actor’s feelings include something not in the provided list, add it under `other_emotions`, **NOT** emotions. (Example: “Frustration” is not in the list, so it must go into `other_emotions`, not `emotions`.)
 - emotions must contain **only** items from the provided list. Do **NOT** include unlisted emotions.
 - `other_emotions` must contain only new, unlisted emotions, and must not duplicate any emotions already listed in `emotions`.
- Each intensity question (Q2, Q4, Q8, Q12) must include all emotions selected in the previous question, including any listed in `other_emotions`.
- Do not include commentary, apologies, or explanations.
- Output **ONLY valid JSON** that strictly follows the provided schema.
- Ensure all strings are properly quoted and escaped.
- End all string values and object properties correctly.

Now, fill in all answers (Q1-Q16) for the given behavior and actor.

A.4 Model Configuration

All experiments were run on Colab Pro. The default temperature was set to 0 for all models to ensure deterministic outputs. However, GPT-5.2 does not permit temperature adjustment when high reasoning is enabled, defaulting to temperature 1. For GPT-OSS, temperature 0 produced response loops on approximately 20% of inputs, resulting in substantial data loss; we therefore set temperature to 1, consistent with the recommended setting in the official documentation.* We share the model responses at temperature 0 in our GitHub repository for reference.

For closed-source models, we report the cost of running the metanorm prediction task in Table A3. High reasoning was enabled for GPT-5.2, high thinking for Gemini-2.5-Pro, and high effort without extended thinking for Claude-4.5-Opus. Maximum token limits were chosen conservatively to avoid mid-response truncation and therefore vary across models according to their generation behaviors.

Each of the 450 scenarios was evaluated with both male and female actor variants, yielding 902 inputs per model. After filtering to only those scenarios with corresponding human annotations, the final evaluation set comprised 853 inputs across 450 unique scenarios per each model.

For Gemini, safety filters were disabled to ensure responses were returned for all hypothetical scenarios. Specifically, all relevant harm categories were set to BLOCK_NONE, and the prompt explicitly stated that the task was for academic research purposes and that scenarios were hypothetical and potentially sensitive. All model outputs were post-processed to ensure valid JSON responses.

All open-source models were run using the llama.cpp inference engine with GGUF-formatted checkpoints obtained from Hugging Face. All layers were placed on GPU, and inference was performed on a single 40 GB A100 GPU. Under quantization, the memory requirements of all open-source models fit within this hardware constraint. Hyperparameter settings for open-source models are reported in Table A4.

Model	Temp.	Max Tokens	API Cost
GPT-5.2	1.0	8,192	\$19.3
Claude-4.5-Opus	0.0	2,048	\$31.8
Gemini-3-Pro	0.0	20,000	\$29.3

Table A3: Hyperparameter settings and API cost estimates for the metanorm prediction task using closed-source models. High reasoning was enabled for GPT-5.2, high thinking for Gemini-2.5-Pro, and high effort without extended thinking for Claude-4.5-Opus. For Gemini, safety filters were disabled by setting all relevant harm categories to BLOCK_NONE.

Model	Temp.	Max Tokens	Context
GPT-OSS-20B	1.0	4,112	8,192
Llama-4-Scout-17B	0.0	1,024	8,192
Gemma-3-12B	0.0	1,024	8,192

Table A4: Hyperparameter settings for the metanorm prediction task using open-source models. All models were run in Colab Pro using the llama.cpp inference engine with GGUF checkpoints, with all layers placed on GPU and deployed on a single 40GB A100.

*<https://github.com/openai/gpt-oss>

A.5 Human Agreement

We estimated Gwet’s AC1/2 scores and intra-class correlation coefficient (ICC) for measuring the degree of agreement among human raters and among LLM models.

Emotion	Violator	Strong Tie	Weak Tie	Stranger
Shame	0.212	0.096	0.045	0.426
Guilt	0.266	0.132	0.448	0.472
Embarrassment	0.218	0.079	0.059	0.045
Contempt	0.139	0.130	0.044	0.058
Disgust	0.219	0.295	0.276	0.227
Anger	0.391	0.177	0.168	0.194
Pride	0.265	0.429	0.440	0.440
Compassion	0.424	0.273	0.305	0.277

Table A5: Inter-rater reliability (ICC) for human judgments of emotion presence across social roles, measuring agreement on whether a given emotion is expressed.

ICC on Actions (Would)			
Action	Acquaintance	Friend	Stranger
Do nothing	0.093	0.124	0.114
Stay away	0.140	0.419	0.054
Gossip	0.006	0.051	0.413
Verbal confrontation	0.137	0.110	0.434
Physical confrontation	0.477	0.451	0.451
Inform authority	0.383	0.364	0.370
Support	0.415	0.364	0.430
Praise	0.405	0.406	0.488

ICC on Actions (Should)			
Action	Acquaintance	Friend	Stranger
Do nothing	0.169	0.136	0.181
Stay away	0.145	0.436	0.099
Gossip	0.809	0.444	0.472
Verbal confrontation	0.158	0.169	0.423
Physical confrontation	0.500	0.491	0.462
Inform authority	0.411	0.389	0.419
Support	0.401	0.279	0.414
Praise	0.340	0.348	0.461

Table A6: **Inter-rater agreement among human raters for action annotations using ICC.** The top panel reports agreement for *would* responses (what individuals would do), and the bottom panel reports agreement for *should* responses (what individuals believe should be done), across weak ties, strong ties, and strangers.

Gwet's AC1 on Emotion Presence				
Emotion	Violator	Strong Tie	Weak Tie	Stranger
Shame	0.227	0.377	0.624	0.779
Guilt	0.224	0.786	0.872	0.921
Embarrassment	0.242	0.131	0.393	0.600
Contempt	0.659	0.465	0.343	0.352
Disgust	0.599	0.274	0.253	0.218
Anger	0.638	0.355	0.481	0.503
Pride	0.609	0.900	0.918	0.938
Compassion	0.881	0.536	0.641	0.656

Gwet's AC2 on Emotion Intensity				
Emotion	Violator	Strong Tie	Weak Tie	Stranger
Shame	0.286	0.259	0.249	0.234
Guilt	0.324	0.253	0.302	0.194
Embarrassment	0.313	0.273	0.306	0.262
Contempt	0.391	0.338	0.339	0.309
Disgust	0.235	0.309	0.245	0.243
Anger	0.423	0.294	0.309	0.220
Pride	0.295	0.337	0.082	0.084
Compassion	0.536	0.412	0.534	0.670

Table A7: **Inter-rater agreement among human raters for emotion annotations using Gwet's coefficients.** The top panel reports Gwet's AC1 for binary emotion presence (whether an emotion is expressed), while the bottom reports Gwet's AC2 for ordinal emotion intensity (rated on a Likert scale of 1-4) conditional on the emotion being present.

Gwet's AC1 on Actions (Would)			
Action	Weak Tie	Strong Tie	Stranger
Do nothing	0.182	0.517	0.167
Stay away	0.701	0.871	0.655
Gossip	0.422	0.545	0.791
Verbal confrontation	0.769	0.286	0.857
Physical confrontation	0.971	0.944	0.975
Inform authority	0.952	0.961	0.944
Support	0.868	0.755	0.954
Praise	0.979	0.973	0.987

Gwet's AC1 on Actions (Should)			
Action	Weak Tie	Strong Tie	Stranger
Do nothing	0.144	0.400	0.269
Stay away	0.721	0.878	0.681
Gossip	0.900	0.922	0.944
Verbal confrontation	0.627	0.219	0.823
Physical confrontation	0.966	0.941	0.985
Inform authority	0.931	0.941	0.925
Support	0.827	0.704	0.931
Praise	0.984	0.973	0.987

Table A8: **Inter-rater agreement for action annotations using Gwet's AC1.** The top panel reports agreement for *would* responses (what individuals would do), and the bottom panel reports agreement for *should* responses (what individuals believe should be done), across weak ties, strong ties, and strangers.

A.6 Human Judgments

A.6.1 No gender differences

Ratings for social inappropriateness and moral wrongness did not differ by violator gender (appropriateness: $t[2609.2] = -1.29, p = .197$; wrongness: $t[2609.8] = -0.59, p = .552$). Distribution of emotional responses did not differ by violator gender for any observer type (actor: $\chi^2(7) = 6.39, p = .495$; strong tie: $\chi^2(7) = 8.42, p = .297$; weak tie: $\chi^2(7) = 2.72, p = .910$; stranger: $\chi^2(7) = 6.68, p = .462$). Similarly, we found no significant difference of violator gender on behavioral responses, both what others would do (strong tie: $\chi^2(7) = 9.06, p = .248$; weak tie: $\chi^2(7) = 6.82, p = .448$; stranger: $\chi^2(7) = 5.84, p = .558$) and what an observer should do (strong tie: $\chi^2(7) = 6.26, p = .509$; weak tie: $\chi^2(7) = 11.01, p = .138$). except for stranger ($\chi^2(7) = 16.54, p = .021$).

A.6.2 Moral dimensions

The distribution of emotions differed significantly across moral foundations for every social target: actor, $\chi^2(28) = 198.00, p < .001$; strong tie, $\chi^2(28) = 164.00, p < .001$; weak tie, $\chi^2(28) = 143.00, p < .001$; and stranger, $\chi^2(28) = 122.00, p < .001$. Descriptively, for the actor target, self-focused emotions were still prominent across foundations, with guilt, shame, and embarrassment generally among the most frequent responses. For example, guilt was especially common for fairness-cheating (24.5%), loyalty-betrayal (22.2%), and care-harm (22.2%), whereas sanctity-degradation showed relatively elevated embarrassment (24.3%) and shame (23.7%). For the strong-tie target, the distribution shifted somewhat away from guilt and toward embarrassment, disgust, and anger, with compassion also relatively elevated in some foundations such as authority-subversion (15.5%) and loyalty-betrayal (16.0%). For weak ties and strangers, other-condemning emotions became more prominent: disgust and contempt were especially common across several foundations, particularly for sanctity-degradation and fairness-cheating, while guilt and pride remained comparatively rare. Overall, the human data suggest that moral foundation systematically shapes emotion attribution, and that this pattern also varies by relational distance, with actor judgments emphasizing self-conscious emotions and more distant targets eliciting more condemning emotions.

The behavioral response distributions differed significantly across moral foundations in all relational contexts, including stranger, weak-tie, and strong-tie targets and both *would* and *should* judgments. Specifically, significant effects were observed for stranger *should* ($\chi^2(28) = 131.0, p < .001$), stranger *would* ($\chi^2(28) = 131.0, p < .001$), weak-tie *should* ($\chi^2(28) = 125.0, p < .001$), weak-tie *would* ($\chi^2(28) = 81.2, p < .001$), strong-tie *should* ($\chi^2(28) = 87.7, p < .001$), and strong-tie *would* ($\chi^2(28) = 101.0, p < .001$). These results indicate that the moral content of a scenario systematically shapes how individuals believe others would and should respond across all relational contexts. Descriptively, Loyalty-betrayal and authority-subversion elicited the highest levels of inaction toward strangers, suggesting that these violations are less likely to prompt intervention from socially distant others. In contrast, care-harm and fairness-cheating showed stronger shifts toward direct engagement in close relationships, with particularly high levels of verbal confrontation among strong ties. Sanctity-degradation was distinguished by a comparatively stronger avoidance profile, with elevated endorsement of Stay away in both stranger and weak-tie contexts.

Together, although these results indicate statistically reliable variation of behavioral and emotional response across moral foundations, these differences were comparatively modest relative to the effects of social distance (i.e., strong tie, weak tie and strangers). We therefore focus our subsequent analyses on the more pronounced variation by social distance.

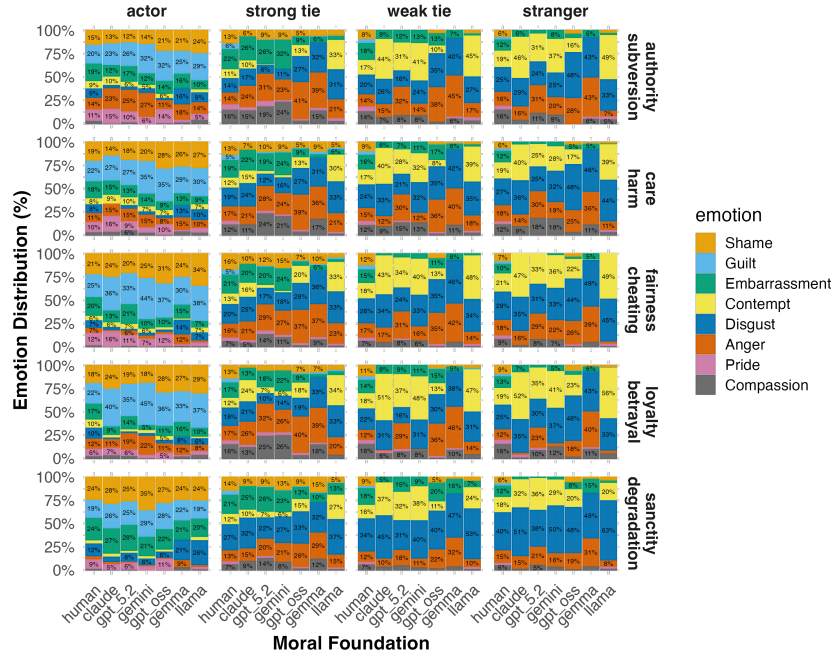


Figure A1: **Emotions Across Moral Foundations for Human and LLMs.** For *actor*, self-focused emotions especially guilt, shame, and embarrassment are most prevalent across foundations. For observers (*strong tie*, *weak tie*, and *stranger*), responses shift toward other-focused emotions, particularly anger, contempt, and disgust. This shift is strongest for sanctity violations, where disgust is especially prominent, and for fairness, loyalty, and authority violations involving socially distant targets, where contempt and disgust increase. Relative to human responses, LLMs place substantially more weight on anger and disgust for observers and much less weight on guilt, shame, and embarrassment, amplifying other-condemning emotions under greater social distance.

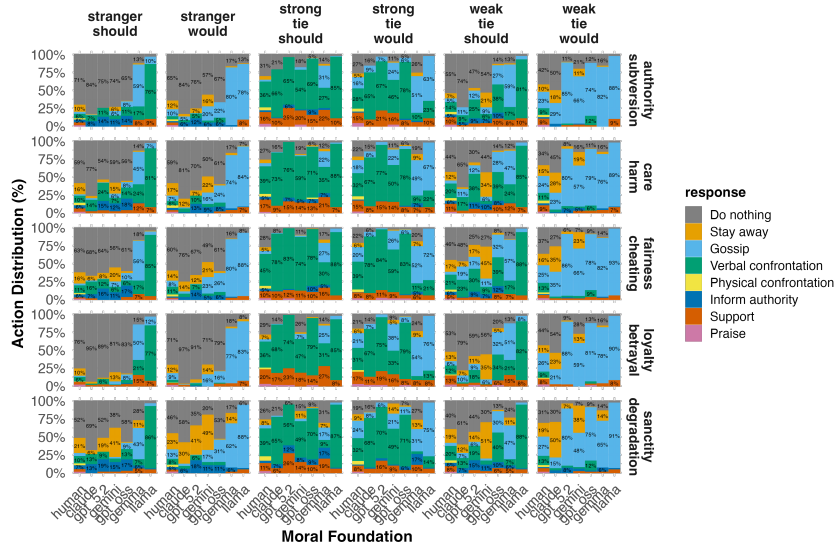


Figure A2: **Behavioral Responses across Moral Foundations by LLMs and Human.** Response distributions vary by moral foundation. Violations involving *authority*, *care*, and *fairness* are more likely to elicit direct intervention, particularly verbal confrontation in strong-tie contexts, whereas *loyalty* violations and *sanctity* violations show more passive or indirect responses, including doing nothing, staying away, and gossiping. These moral-foundation differences are clearest in stranger and weak-tie conditions.

A.6.3 Human Consensus

Category	Actor				Strong Tie				Weak Tie				Stranger			
	Prop	Self	Other	Pos	Prop	Self	Other	Pos	Prop	Self	Other	Pos	Prop	Self	Other	Pos
Panel A: Majority-Vote Consensus																
Both Self + Other	.102	1.0	1.0	1.0	.152	1.0	1.0	0.0	.061	1.0	1.0	0.0	.014	1.0	1.0	0.0
Self-focused	.428	1.0	0.0	1.0	.176	1.0	0.0	0.0	.079	1.0	0.0	0.0	.034	1.0	0.0	0.0
Other-focused	.161	0.0	1.0	0.0	.260	0.0	1.0	0.0	.348	0.0	1.0	0.0	.434	0.0	1.0	0.0
Positive Only	.025	0.0	0.0	1.0	.118	0.0	0.0	1.0	.075	0.0	0.0	1.0	.075	0.0	0.0	1.0
No Emotion	.285	0.0	0.0	0.0	.294	0.0	0.0	0.0	.437	0.0	0.0	0.0	.443	0.0	0.0	0.0
Panel B: At-Least-Two Agreement																
Both Self + Other	.543	2.0	1.0	1.0	.595	2.0	2.0	0.0	.446	1.0	2.0	0.0	.276	1.0	2.0	0.0
Self-focused	.305	2.0	0.0	1.0	.145	1.0	0.0	0.5	.102	1.0	0.0	0.0	.072	1.0	0.0	1.0
Other-focused	.097	0.0	1.0	0.0	.158	0.0	1.0	0.0	.312	0.0	2.0	0.0	.511	0.0	2.0	0.0
Positive Only	.009	0.0	0.0	1.0	.070	0.0	0.0	1.0	.052	0.0	0.0	1.0	.068	0.0	0.0	1.0
No Emotion	.045	0.0	0.0	0.0	.032	0.0	0.0	0.0	.088	0.0	0.0	0.0	.072	0.0	0.0	0.0

Table A9: Distribution of stimuli based on the type of emotions expressed (e.g., only self-focused, only other-focused, both self- and other-focused, positive only, or no emotion) across social ties under majority-vote and fixed-threshold (≥ 2 raters agree) consensus. Columns report the proportion (Prop) and the median number of emotions selected for self-focused (Self), other-focused (Other), and positive (Pos) categories. Note that “Self-focused” and “Other-focused” may include instances where positive emotions co-occur with self- or other-directed emotions. Under the fixed-threshold criterion, the number of positive samples for self-regulation (“Both Self + Other” and “Self-focused”) and other-regulation (“Both Self + Other” and “Other-focused”) used for LLM benchmarking is approximately doubled. A similar pattern is observed for the median number of emotions selected.

Category	Strong Tie		Weak Tie		Stranger	
	Would	Should	Would	Should	Would	Should
Panel A: Majority-Vote Consensus						
Sanction	.616	.532	.505	.335	.235	.206
Non-sanction	.384	.468	.495	.665	.765	.794
Panel B: At-Least-Two Agreement						
Sanction	.826	.740	.776	.604	.593	.459
Non-sanction	.174	.260	.224	.396	.407	.541

Table A10: Distribution of stimuli based on the type of action taken under majority-vote and fixed-threshold (≥ 2 raters agree) consensus. Sanction actions include *stay away*, *gossip*, *verbal/physical confrontation*, and *inform authorities*, while non-sanction actions include *do nothing*, *support*, and *praise*. Overall, sanctioning behavior increases substantially under the fixed-threshold criterion and is more common for stronger social ties, with a consistent decline from friends to strangers.

A.7 Model Evaluation - Additional Results

A.7.1 Norm Appropriateness

We further compared model ratings of perceived wrongness and inappropriateness with human rating using ordinal logistic regression (see Table A11). Model identity significantly predicted both outcomes relative to human responses. Across both measures, Gemma exhibited the largest increase in perceived severity (wrongness: $b = 0.841$, OR = 2.319; inappropriateness: $b = 0.803$, OR = 2.232), followed by LLaMA (wrongness: $b = 0.601$, OR = 1.823; inappropriateness: $b = 0.548$, OR = 1.730) and GPT-OSS (wrongness: $b = 0.611$, OR = 1.843; inappropriateness: $b = 0.452$, OR = 1.571), all $p < .001$. In contrast, Claude was associated with significantly lower ratings than the human baseline on both dimensions (wrongness: $b = -0.611$, OR = 0.543; inappropriateness: $b = -0.435$, OR = 0.647; both $p < .001$). GPT-5.2 and Gemini did not differ from humans in perceived wrongness (GPT-5.2: $b = 0.009$, OR = 1.009, $p = .486$; Gemini: $b = -0.003$, OR = 0.997, $p = .797$), yet they tend to judge norm violations as more inappropriate (GPT-5.2: $b = 0.266$, OR = 1.305; Gemini: $b = 0.315$, OR = 1.371; both $p < .001$). Overall, the pattern suggest that models tend to over predict the wrongness and inappropriateness of norm violation than human (cf. Claude) especially for open sourced model.

Model (vs. Human)	Wrongness					Inappropriateness				
	Estimate	SE	t	p	OR	Estimate	SE	t	p	OR
Claude	-0.611	0.012	-49.85	< .001	0.543	-0.435	0.012	-35.7	< .001	0.647
GPT-5.2	0.009	0.013	0.70	.486	1.009	0.266	0.013	21.3	< .001	1.305
Gemini	-0.003	0.013	-0.26	.797	0.997	0.315	0.013	24.7	< .001	1.371
GPT-OSS	0.611	0.013	47.55	< .001	1.843	0.452	0.013	35.8	< .001	1.571
Gemma	0.841	0.012	69.98	< .001	2.319	0.803	0.012	66.0	< .001	2.232
LLaMA	0.601	0.012	50.37	< .001	1.823	0.548	0.012	45.5	< .001	1.730

Table A11: Ordinal logistic regression results predicting perceived wrongness and inappropriateness ratings. Human responses served as the reference category. Coefficients represent log-odds differences in the likelihood of assigning higher ratings relative to human responses. Odds ratios (OR) greater than 1 indicate increased likelihood of higher ratings, whereas values below 1 indicate decreased likelihood. p -values are based on Wald tests.

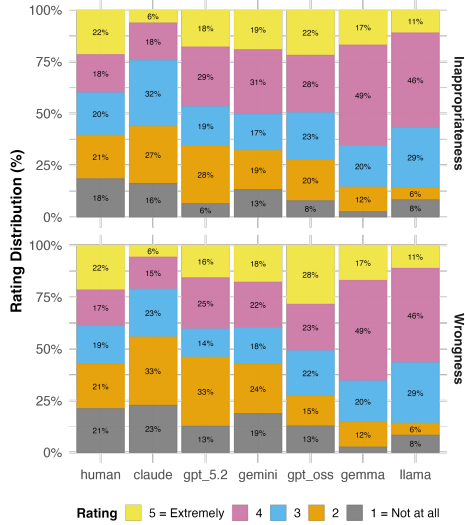


Figure A3: **Norm judgments** Distribution of social inappropriateness and wrongness scores reported by human participants and LLMs. Human responses are relatively evenly spread across the scale, while LLMs responses vary in their concentration.

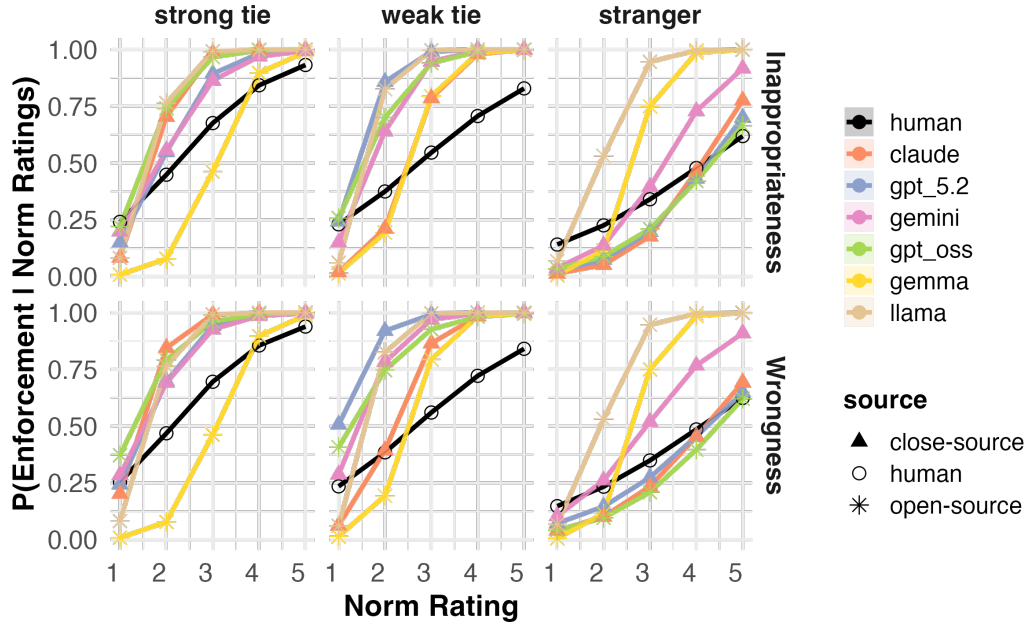


Figure A4: **Predicted probability of enforcement (vs. non-enforcement) as a function of perceived inappropriateness and wrongness across social relationships.** Enforcement responses are defined as actions that impose social or reputational costs, including gossip, informing an authority, verbal or physical confrontation, and social ostracism (staying away), whereas non-enforcement responses include inaction (doing nothing), praise, and support for the norm violator. Emotion strength varies from 0=not at all to 4=extremely. In general, higher ratings of inappropriateness and wrongness predict a greater likelihood of enforcement, although the magnitude and shape of this relationship differ across social relationships and sources. For human judgments, the increase in enforcement is more gradual and strongly moderated by social distance: predicted enforcement is highest for strong ties, somewhat lower for weak ties, and lowest for strangers across the rating scale. By contrast, several model responses increase more sharply and often reach near-ceiling probabilities at moderate levels of perceived inappropriateness or wrongness, particularly in strong-tie and weak-tie conditions. Shaded areas represent 95% confidence intervals.

A.7.2 Emotion and Action Profiles

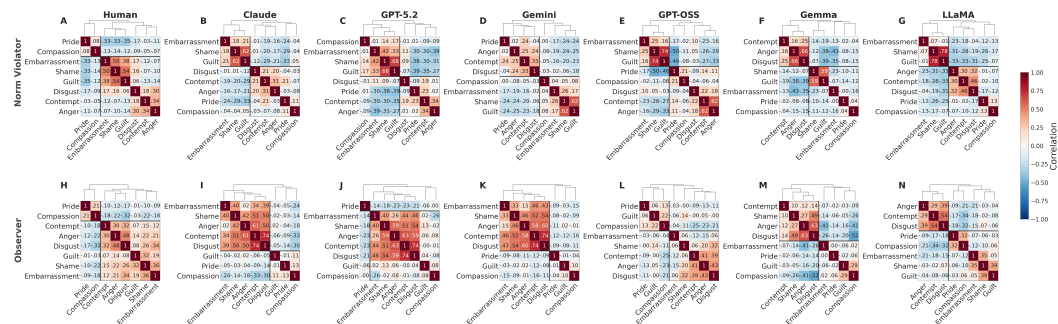


Figure A5: **Pairwise Spearman correlations between emotion intensities.** Observer emotions are aggregated as the mean of non-zero strengths across observer types. LLMs broadly replicate the human co-occurrence structure of emotions for norm violators, while for observer emotions, all LLMs, except LLaMa, restructure the clustering: shame aligns with condemnation emotions while guilt pairs with compassion. This suggests that unlike humans most LLMs treat shame and guilt as functionally distinct in observers: shame as accusatory, guilt as signaling remorse.

	Norm violator		Observer		
	Men	Women	Strong	Weak	Stranger
Panel A. Human					
Shame	20.1% (549)	18.9% (520)	14.1% (757)	9.9% (432)	6.8% (265)
Guilt	21.8% (597)	21.6% (594)	5.0% (269)	3.6% (159)	2.6% (100)
Embarrassment	20.0% (547)	19.6% (539)	19.7% (1062)	16.3% (716)	11.9% (465)
Contempt	6.8% (185)	8.2% (226)	11.9% (640)	17.1% (750)	19.0% (740)
Disgust	9.6% (263)	9.3% (255)	19.9% (1070)	25.3% (1111)	29.7% (1156)
Anger	9.3% (256)	10.0% (276)	15.4% (827)	14.2% (624)	16.1% (627)
Pride	9.7% (265)	9.4% (259)	2.8% (149)	2.6% (113)	2.2% (86)
Compassion	2.8% (76)	3.2% (87)	11.2% (604)	10.9% (479)	11.8% (458)
Panel B. Claude					
Shame	13.4% (119)	12.7% (121)	4.4% (122)	0.0% (0)	0.0% (0)
Guilt	19.7% (175)	19.0% (181)	0.0% (1)	0.0% (0)	0.0% (0)
Embarrassment	9.2% (82)	9.8% (93)	13.6% (379)	3.7% (68)	1.9% (25)
Contempt	3.7% (33)	4.6% (44)	9.7% (270)	29.8% (545)	30.9% (404)
Disgust	0.9% (8)	1.3% (12)	15.4% (427)	24.2% (443)	28.4% (372)
Anger	6.5% (58)	6.2% (59)	13.6% (377)	8.2% (150)	10.1% (132)
Pride	7.7% (68)	6.8% (65)	0.5% (14)	0.1% (2)	0.2% (2)
Compassion	0.9% (8)	1.6% (15)	6.6% (183)	4.2% (76)	2.4% (32)
Panel C. GPT-5.2					
Shame	15.7% (175)	14.3% (165)	7.9% (216)	0.0% (0)	0.0% (0)
Guilt	22.4% (249)	23.2% (268)	0.7% (20)	0.0% (0)	0.0% (0)
Embarrassment	13.4% (149)	15.8% (183)	18.9% (519)	8.3% (187)	1.0% (16)
Contempt	5.0% (56)	4.3% (50)	4.5% (122)	28.7% (648)	29.7% (493)
Disgust	2.2% (25)	2.2% (25)	12.1% (332)	19.9% (449)	28.2% (467)
Anger	10.2% (113)	10.1% (117)	24.0% (657)	24.7% (557)	25.2% (418)
Pride	7.2% (80)	5.8% (67)	1.9% (53)	0.3% (6)	0.1% (2)
Compassion	2.0% (22)	2.3% (27)	16.6% (455)	7.9% (178)	9.8% (163)
Panel D. Gemini					
Shame	17.7% (151)	15.8% (140)	8.5% (186)	0.0% (0)	0.0% (0)
Guilt	28.0% (238)	26.7% (236)	0.4% (8)	0.0% (0)	0.0% (0)
Embarrassment	9.0% (77)	9.5% (84)	20.0% (436)	8.4% (136)	3.3% (40)
Contempt	3.5% (30)	4.1% (36)	3.2% (69)	34.9% (565)	30.6% (371)
Disgust	2.0% (17)	2.4% (21)	14.5% (316)	28.9% (468)	33.3% (403)
Anger	9.4% (80)	10.4% (92)	19.6% (426)	10.3% (167)	16.0% (194)
Pride	3.5% (30)	3.6% (32)	0.5% (11)	0.0% (0)	0.0% (0)
Compassion	0.7% (6)	0.8% (7)	13.7% (298)	6.7% (108)	8.8% (107)
Panel E. GPT-OSS					
Shame	26.0% (217)	27.2% (229)	6.8% (129)	3.1% (50)	3.3% (36)
Guilt	33.7% (281)	32.3% (272)	0.8% (15)	0.2% (3)	0.1% (1)
Embarrassment	13.5% (113)	12.6% (106)	9.1% (173)	15.3% (246)	4.9% (54)
Contempt	5.2% (43)	4.6% (39)	15.6% (297)	11.0% (177)	19.8% (219)
Disgust	2.3% (19)	2.4% (20)	26.9% (512)	35.2% (567)	47.2% (522)
Anger	6.4% (53)	7.4% (62)	36.4% (692)	33.0% (532)	23.2% (256)
Pride	10.2% (85)	10.5% (88)	0.6% (12)	0.4% (6)	0.1% (1)
Compassion	0.7% (6)	0.7% (6)	3.4% (65)	1.6% (25)	1.3% (14)
Panel F. Gemma					
Shame	24.6% (225)	24.3% (223)	9.5% (166)	0.4% (5)	0.4% (5)
Guilt	27.1% (248)	28.1% (258)	1.5% (26)	0.1% (2)	0.1% (2)
Embarrassment	14.7% (135)	17.6% (162)	5.7% (99)	8.1% (114)	7.7% (103)
Contempt	1.2% (11)	0.9% (8)	0.2% (3)	0.5% (7)	0.7% (10)
Disgust	14.8% (136)	14.4% (132)	32.7% (571)	42.2% (596)	45.4% (610)
Anger	14.6% (134)	11.3% (104)	35.5% (619)	40.5% (572)	37.4% (502)
Pride	0.4% (4)	0.5% (5)	0.5% (8)	0.2% (3)	0.2% (3)
Compassion	2.5% (23)	2.9% (27)	13.4% (234)	6.9% (97)	7.0% (94)
Panel G. LLaMA					
Shame	29.3% (237)	26.3% (209)	4.5% (99)	1.7% (25)	1.4% (16)
Guilt	31.9% (258)	29.4% (234)	1.2% (26)	0.3% (4)	0.1% (1)
Embarrassment	9.9% (80)	12.2% (97)	5.9% (128)	2.6% (37)	1.1% (13)
Contempt	5.1% (41)	5.2% (41)	31.2% (681)	40.7% (586)	42.0% (480)
Disgust	11.6% (94)	12.2% (97)	33.3% (725)	36.1% (520)	44.0% (503)
Anger	6.7% (54)	8.2% (65)	20.3% (443)	14.7% (212)	7.0% (80)
Pride	3.1% (25)	3.1% (25)	0.6% (14)	0.9% (13)	1.0% (11)
Compassion	2.3% (19)	3.5% (28)	2.5% (55)	2.9% (42)	3.5% (40)

Decrease Increase Non-monotonic Low throughput (<3%)

Table A12: Major 8 emotion distributions across humans and LLMs. Row shading reflects strictly monotonic trends across observer social distance (Strong → Weak → Stranger).

Reaction type	Strong	Weak	Stranger	Overall
Panel A. Human				
Do nothing	580 (22.2%)	986 (37.7%)	1,576 (60.3%)	3,142 (40.1%)
Stay away	162 (6.2%)	374 (14.3%)	407 (15.6%)	943 (12.0%)
Gossip	519 (19.9%)	646 (24.7%)	251 (9.6%)	1,416 (18.1%)
Verbal confrontation	848 (32.5%)	287 (11.0%)	175 (6.7%)	1,310 (16.7%)
Physical confrontation	70 (2.7%)	38 (1.5%)	35 (1.3%)	143 (1.8%)
Inform authority	58 (2.2%)	73 (2.8%)	86 (3.3%)	217 (2.8%)
Support	334 (12.8%)	182 (7.0%)	65 (2.5%)	581 (7.4%)
Praise	42 (1.6%)	27 (1.0%)	18 (0.7%)	87 (1.1%)
Panel B. Claude				
Do nothing	125 (14.7%)	376 (44.1%)	675 (79.1%)	1,176 (46.0%)
Stay away	1 (0.1%)	249 (29.2%)	105 (12.3%)	355 (13.9%)
Gossip	60 (7.0%)	198 (23.2%)	25 (2.9%)	283 (11.1%)
Verbal confrontation	579 (67.9%)	9 (1.1%)	17 (2.0%)	605 (23.6%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	13 (1.5%)	16 (1.9%)	29 (3.4%)	58 (2.3%)
Support	73 (8.6%)	5 (0.6%)	2 (0.2%)	80 (3.1%)
Praise	2 (0.2%)	0 (0.0%)	0 (0.0%)	2 (0.1%)
Panel C. GPT-5.2				
Do nothing	20 (2.3%)	72 (8.4%)	589 (69.1%)	681 (26.6%)
Stay away	0 (0.0%)	28 (3.3%)	115 (13.5%)	143 (5.6%)
Gossip	34 (4.0%)	707 (82.9%)	4 (0.5%)	745 (29.1%)
Verbal confrontation	626 (73.4%)	7 (0.8%)	45 (5.3%)	678 (26.5%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	17 (2.0%)	34 (4.0%)	100 (11.7%)	151 (5.9%)
Support	152 (17.8%)	5 (0.6%)	0 (0.0%)	157 (6.1%)
Praise	4 (0.5%)	0 (0.0%)	0 (0.0%)	4 (0.2%)
Panel D. Gemini				
Do nothing	68 (8.0%)	153 (17.9%)	423 (49.6%)	644 (25.2%)
Stay away	41 (4.8%)	180 (21.1%)	214 (25.1%)	435 (17.0%)
Gossip	197 (23.1%)	480 (56.3%)	148 (17.4%)	825 (32.2%)
Verbal confrontation	413 (48.4%)	4 (0.5%)	16 (1.9%)	433 (16.9%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	11 (1.3%)	26 (3.0%)	52 (6.1%)	89 (3.5%)
Support	120 (14.1%)	10 (1.2%)	0 (0.0%)	130 (5.1%)
Praise	3 (0.4%)	0 (0.0%)	0 (0.0%)	3 (0.1%)
Panel E. GPT-OSS				
Do nothing	47 (5.5%)	98 (11.5%)	564 (66.1%)	709 (27.7%)
Stay away	8 (0.9%)	3 (0.4%)	10 (1.2%)	21 (0.8%)
Gossip	61 (7.2%)	657 (77.0%)	191 (22.4%)	909 (35.5%)
Verbal confrontation	655 (76.8%)	68 (8.0%)	27 (3.2%)	750 (29.3%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	26 (3.0%)	11 (1.3%)	54 (6.3%)	91 (3.6%)
Support	54 (6.3%)	15 (1.8%)	7 (0.8%)	76 (3.0%)
Praise	2 (0.2%)	1 (0.1%)	0 (0.0%)	3 (0.1%)
Panel F. Gemma				
Do nothing	188 (22.0%)	130 (15.2%)	145 (17.0%)	463 (18.1%)
Stay away	67 (7.9%)	37 (4.3%)	37 (4.3%)	141 (5.5%)
Gossip	409 (47.9%)	651 (76.3%)	635 (74.4%)	1,695 (66.2%)
Verbal confrontation	60 (7.0%)	0 (0.0%)	0 (0.0%)	60 (2.3%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	66 (7.7%)	20 (2.3%)	21 (2.5%)	107 (4.2%)
Support	63 (7.4%)	15 (1.8%)	15 (1.8%)	93 (3.6%)
Praise	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Panel G. LLaMA				
Do nothing	24 (2.8%)	24 (2.8%)	55 (6.4%)	103 (4.0%)
Stay away	0 (0.0%)	0 (0.0%)	7 (0.8%)	7 (0.3%)
Gossip	619 (72.6%)	773 (90.6%)	738 (86.5%)	2,130 (83.2%)
Verbal confrontation	148 (17.4%)	0 (0.0%)	0 (0.0%)	148 (5.8%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	7 (0.8%)	4 (0.5%)	4 (0.5%)	15 (0.6%)
Support	55 (6.4%)	52 (6.1%)	49 (5.7%)	156 (6.1%)
Praise	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Decrease Increase Non-monotonic Low throughput (<3%)

Table A13: Distribution of behavioral reactions (would) across humans and LLMs. Row shading reflects trends across social distance (Strong → Weak → Stranger).

Reaction type	Strong	Weak	Stranger	Overall
Panel A. Human				
Do nothing	734 (28.1%)	1,236 (47.3%)	1,678 (64.2%)	3,648 (46.5%)
Stay away	154 (5.9%)	355 (13.6%)	379 (14.5%)	888 (11.3%)
Gossip	100 (3.8%)	125 (4.8%)	68 (2.6%)	293 (3.7%)
Verbal confrontation	1,016 (38.9%)	459 (17.6%)	228 (8.7%)	1,703 (21.7%)
Physical confrontation	74 (2.8%)	45 (1.7%)	23 (0.9%)	142 (1.8%)
Inform authority	101 (3.9%)	123 (4.7%)	125 (4.8%)	349 (4.5%)
Support	394 (15.1%)	247 (9.5%)	93 (3.6%)	734 (9.4%)
Praise	40 (1.5%)	23 (0.9%)	19 (0.7%)	82 (1.0%)
Panel B. Claude				
Do nothing	151 (17.7%)	571 (66.9%)	675 (79.1%)	1,397 (54.6%)
Stay away	0 (0.0%)	44 (5.2%)	28 (3.3%)	72 (2.8%)
Gossip	5 (0.6%)	92 (10.8%)	7 (0.8%)	104 (4.1%)
Verbal confrontation	583 (68.3%)	107 (12.5%)	90 (10.6%)	780 (30.5%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	24 (2.8%)	27 (3.2%)	47 (5.5%)	98 (3.8%)
Support	88 (10.3%)	12 (1.4%)	6 (0.7%)	106 (4.1%)
Praise	2 (0.2%)	0 (0.0%)	0 (0.0%)	2 (0.1%)
Panel C. GPT-5.2				
Do nothing	28 (3.3%)	377 (44.2%)	578 (67.8%)	983 (38.4%)
Stay away	0 (0.0%)	76 (8.9%)	55 (6.4%)	131 (5.1%)
Gossip	0 (0.0%)	89 (10.4%)	2 (0.2%)	91 (3.6%)
Verbal confrontation	592 (69.4%)	206 (24.2%)	98 (11.5%)	896 (35.0%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	49 (5.7%)	77 (9.0%)	110 (12.9%)	236 (9.2%)
Support	181 (21.2%)	27 (3.2%)	10 (1.2%)	218 (8.5%)
Praise	3 (0.4%)	1 (0.1%)	0 (0.0%)	4 (0.2%)
Panel D. Gemini				
Do nothing	149 (17.5%)	363 (42.6%)	518 (60.7%)	1,030 (40.3%)
Stay away	28 (3.3%)	320 (37.5%)	174 (20.4%)	522 (20.4%)
Gossip	26 (3.0%)	55 (6.4%)	41 (4.8%)	122 (4.8%)
Verbal confrontation	481 (56.4%)	38 (4.5%)	34 (4.0%)	553 (21.6%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	29 (3.4%)	59 (6.9%)	80 (9.4%)	168 (6.6%)
Support	138 (16.2%)	18 (2.1%)	6 (0.7%)	162 (6.3%)
Praise	2 (0.2%)	0 (0.0%)	0 (0.0%)	2 (0.1%)
Panel E. GPT-OSS				
Do nothing	53 (6.2%)	121 (14.2%)	564 (66.1%)	738 (28.8%)
Stay away	9 (1.1%)	21 (2.5%)	12 (1.4%)	42 (1.6%)
Gossip	3 (0.4%)	252 (29.5%)	68 (8.0%)	323 (12.6%)
Verbal confrontation	621 (72.8%)	321 (37.6%)	76 (8.9%)	1,018 (39.8%)
Physical confrontation	1 (0.1%)	0 (0.0%)	0 (0.0%)	1 (0.0%)
Inform authority	61 (7.2%)	69 (8.1%)	116 (13.6%)	246 (9.6%)
Support	104 (12.2%)	68 (8.0%)	17 (2.0%)	189 (7.4%)
Praise	1 (0.1%)	1 (0.1%)	0 (0.0%)	2 (0.1%)
Panel F. Gemma				
Do nothing	146 (17.1%)	133 (15.6%)	147 (17.2%)	426 (16.6%)
Stay away	31 (3.6%)	24 (2.8%)	24 (2.8%)	79 (3.1%)
Gossip	215 (25.2%)	447 (52.4%)	432 (50.6%)	1,094 (42.8%)
Verbal confrontation	223 (26.1%)	150 (17.6%)	150 (17.6%)	523 (20.4%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	67 (7.9%)	21 (2.5%)	22 (2.6%)	110 (4.3%)
Support	171 (20.0%)	78 (9.1%)	78 (9.1%)	327 (12.8%)
Praise	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Panel G. LLaMA				
Do nothing	24 (2.8%)	24 (2.8%)	31 (3.6%)	79 (3.1%)
Stay away	0 (0.0%)	0 (0.0%)	1 (0.1%)	1 (0.0%)
Gossip	20 (2.3%)	31 (3.6%)	48 (5.6%)	99 (3.9%)
Verbal confrontation	747 (87.6%)	738 (86.5%)	716 (83.9%)	2,201 (86.0%)
Physical confrontation	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Inform authority	7 (0.8%)	7 (0.8%)	7 (0.8%)	21 (0.8%)
Support	55 (6.4%)	53 (6.2%)	50 (5.9%)	158 (6.2%)
Praise	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Decrease Increase Non-monotonic Low throughput (<3%)

Table A14: Distribution of ideal behavioral reactions (should) across humans and language models. Row shading reflects trends across social distance (Strong → Weak → Stranger).

Model	Violator	Strong Tie	Weak Tie	Stranger
Human	2.10	2.07	1.69	1.50
Claude	1.77	2.23	1.83	1.85
Gemini	1.78	2.24	2.04	1.84
GPT-5.2	2.25	2.84	2.53	2.41
GPT-OSS	2.08	2.31	2.08	1.94
Gemma	2.15	2.04	1.69	1.62
LLaMA	1.95	2.62	1.74	1.39

Table A15: Mean number of major 8 emotions selected per response across social roles. Humans select an average of 2.10 emotions for the violator, which declines steadily to 1.50 for strangers: a 29% drop reflecting increasing emotional uncertainty with social distance. Among LLMs, GPT-5.2 consistently selects the most emotions across all roles (2.25–2.84), suggesting it over-attributes emotional states. Gemma most closely matches human counts across all roles (2.15, 2.04, 1.69, 1.62 vs. human 2.10, 2.07, 1.69, 1.50). Most LLMs maintain higher stranger counts than humans, suggesting they don’t attenuate emotional attribution with social distance as strongly as humans do.

A.7.3 Other Emotions

Model	Other Emotions
Claude	
Violator	Frustration, Indifference, Satisfaction, Relief, Amusement, Anxiety, Excitement, Sadness, Insecurity, Righteousness, Entitlement, Regret, Determination, Hurt, Fear, Jealousy, Curiosity, Conflict, Concern, Contentment, Defensiveness, Justified, Conviction, Resentment, Vindication, Disappointment, Enjoyment, Nervousness, Despair, Happiness, Annoyance, Stress, Loneliness, Confusion, Longing, Self-loathing, Hopelessness, Irritation, Discomfort, Hopefulness, Grief, Resignation, Relaxation, Thrill, Vindictiveness, Worry, Comfort, Defensive, Overwhelm, Distrust, Pleasure, Desperation, Helplessness, Indignation, Arousal, Stubbornness, Protectiveness, Affection, Anticipation, Shock, Betrayal, Distress, Hope, Love, Desire, Vulnerability, Surprise, Tolerance, Acceptance, Possessiveness, Self-justification, Assertiveness, Suspicion, Exhaustion, Weariness, Reluctance, Joy, Resolve, Justification, Serenity
Strong Tie	Disappointment, Concern, Shock, Worry, Amusement, Surprise, Sadness, Disapproval, Confusion, Annoyance, Fear, Frustration, Betrayal, Hurt, Understanding, Discomfort, Sympathy, Curiosity, Horror, Relief, Approval, Mild disapproval, Happiness, Grief, Support, Anxiety, Awkwardness
Weak Tie	Disapproval, Surprise, Annoyance, Shock, Curiosity, Disappointment, Confusion, Discomfort, Concern, Amusement, Fear, Indifference, Sadness, Awkwardness, Frustration, Horror, Distrust, Sympathy, Skepticism, Happiness, Respect, Approval, Suspicion, Pity
Stranger	Disapproval, Shock, Annoyance, Surprise, Confusion, Discomfort, Curiosity, Concern, Fear, Amusement, Indifference, Horror, Mild disapproval, Awkwardness, Offense, Respect, Approval, Skepticism, Irritation, Sadness, Suspicion, Wariness, Disappointment, Outrage, Frustration, Inconvenience, Alarm
GPT-5.2	
Violator	Anxiety, Sadness, Regret, Relief, Frustration, Jealousy, Fear, Excitement, Insecurity, Hurt, Attraction, Resentment, Amusement, Arousal, Entitlement, Curiosity, Concern, Annoyance, Happiness, Worry, Loneliness, Affection, Hopelessness, Despair, Satisfaction, Desire, Indifference, Nausea, Enjoyment, Disappointment, Pleasure, Conflicted, Discomfort, Greed, Stress, Longing, Craving, Grief, Hope, Helplessness, Love, Heartbreak, Hatred, Sexual arousal, Lust, Confusion, Determination, Defiance, Tiredness, Unease, Surprise, Irritation, Defensiveness, Infatuation, Self-loathing, Self-admiration, Remorse, Numbness, Vindictiveness, Nervousness, Overwhelmed, Suspicion, Peacefulness
Strong Tie	Disappointment, Concern, Worry, Sadness, Fear, Hurt, Amusement, Confusion, Disapproval, Shock, Happiness, Betrayal, Relief, Annoyance, Awkwardness, Discomfort
Weak Tie	Disapproval, Concern, Disappointment, Annoyance, Unease, Shock, Confusion, Distrust, Sadness, Fear, Judgment, Amusement, Pity, Awkwardness, Discomfort, Curiosity, Alarm, Worry, Happiness, Suspicion, Admiration, Indifference, Respect, Surprise
Stranger	Concern, Disapproval, Fear, Alarm, Shock, Annoyance, Confusion, Unease, Sadness, Indifference, Irritation, Disappointment, Suspicion, Sympathy, Awkwardness, Distrust, Uncomfortable, Surprise, Skepticism
Gemini	
Violator	Anxiety, Frustration, Fear, Relief, Sadness, Jealousy, Excitement, Amusement, Satisfaction, Regret, Annoyance, Insecurity, Despair, Love, Entitlement, Happiness, Betrayal, Hopelessness, Enjoyment, Desperation, Desire, Hurt, Panic, Helplessness, Resentment, Curiosity, Self-righteousness, Sexual arousal, Loneliness, Disappointment, Pleasure, Exhaustion, Attraction, Determination, Joy, Longing, Possessiveness, Affection, Conflicted, Righteousness, Nervousness, Concern, Discomfort, Shock, Obsession, Hatred, Morbid curiosity, Self-admiration, Confusion, Indignation, Vindication, Arousal, Hopeful, Hope, Rejection, Suspicion, Peaceful
Strong Tie	Disappointment, Concern, Worry, Sadness, Shock, Confusion, Amusement, Annoyance, Fear, Betrayal, Surprise, Frustration, Helplessness, Horror, Relief, Happiness, Curiosity, Excitement, Admiration, Hurt, Support, Resentment
Weak Tie	Disapproval, Confusion, Shock, Annoyance, Pity, Amusement, Surprise, Fear, Curiosity, Distrust, Concern, Unease, Awkwardness, Discomfort, Respect, Suspicion, Envy, Horror, Creeped out, Disappointment, Sadness, Frustration
Stranger	Confusion, Shock, Annoyance, Fear, Amusement, Concern, Surprise, Disapproval, Pity, Discomfort, Unease, Frustration, Horror, Wary, Curiosity, Distrust
GPT-OSS	
Violator	Regret, Sadness, Amusement, Resentment, Satisfaction, Curiosity, Excitement, Arousal, Peaceful, Love, Pleasure, Hopelessness, Remorse, Disappointment, Joy, Despair, Enjoyment, Jealousy, Hurt, Anxiety, Grief
Strong Tie	Disappointment, Sadness, Hurt, Frustration, Concern
Weak Tie	Sadness, Disapproval, Disappointment, Annoyance, Skeptical
Stranger	Curiosity, Sadness
Gemma	
Violator	None
Strong Tie	Concern, Disappointment
Weak Tie	Concern, Disappointment
Stranger	Concern, Disappointment
LLaMA	
Violator	Curiosity
Strong Tie	Concern, Disapproval, Surprise
Weak Tie	None
Stranger	None

Table A16: Unique “other” (non-core) emotions produced by each model across social roles. Emotions are listed in descending order of frequency within each category.

A.7.4 Emotion Intensity Distribution

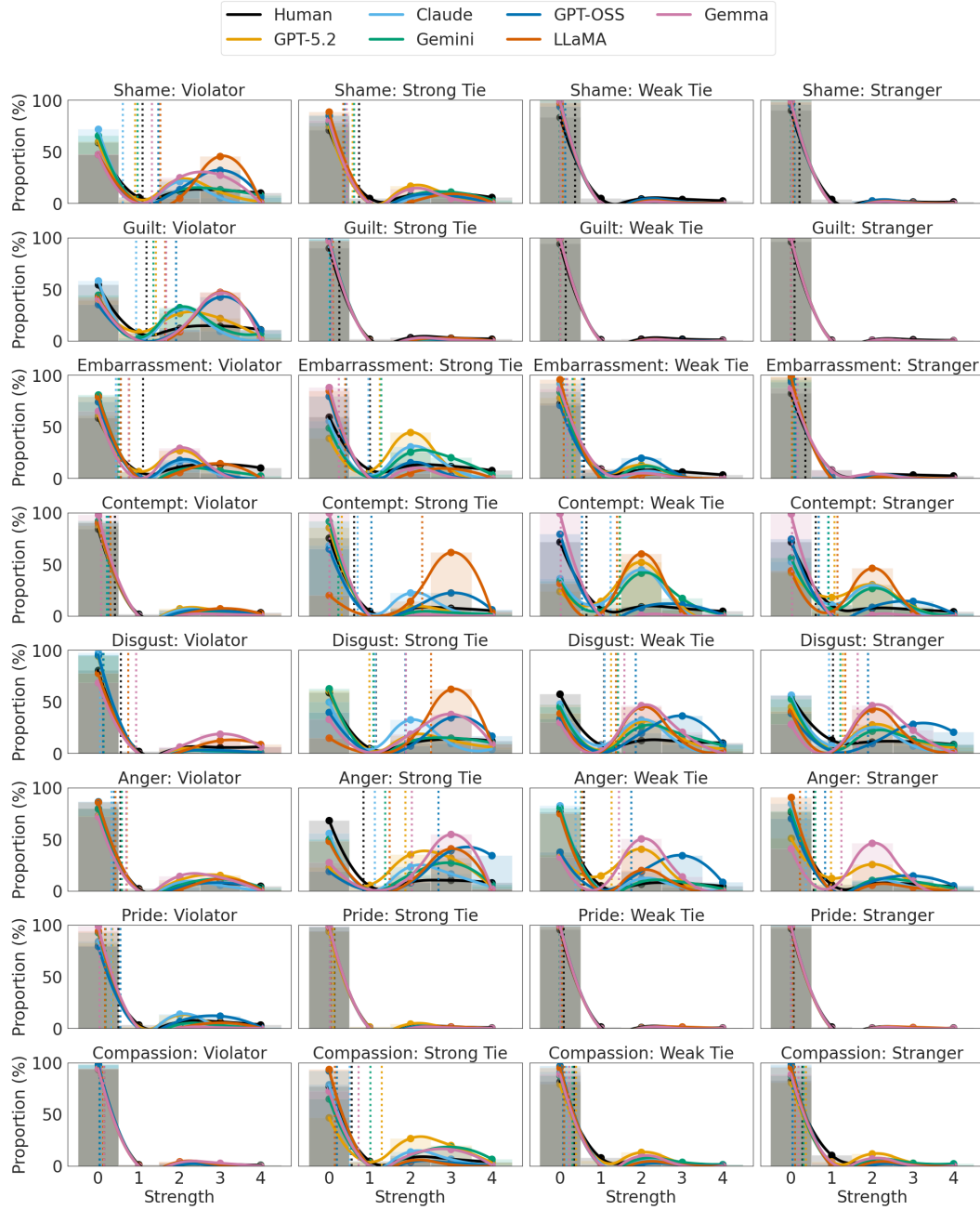


Figure A6: **Distribution of emotion intensity** on a Likert scale of 0 to 4, including instances where the emotion was assigned zero intensity (i.e., not selected), across human annotations and model responses. LLMs show more polarized intensity assignments, concentrating on 2-3 intensity levels, while humans distribute more evenly across the full 1-4 scale. For other-focused emotions in observers, LLMs show a systematic downward shift with social distance: intensities assigned at 3 for strong ties drop to 2 for weak ties and strangers (e.g., GPT-OSS for contempt, LLaMA for contempt, disgust, and anger).

Emotion	Level	Human	GPT-5.2	Claude	Gemini	GPT-OSS	LLaMA	Gemma
Shame	Violator	1.947	1.472	0.959	1.573	1.330	0.639	1.019
	Strong Tie	1.953	1.082	1.262	1.438	1.303	0.903	0.964
	Weak Tie	1.967	–	–	–	1.177	1.535	0.000
	Stranger	1.875	–	–	–	1.225	1.406	0.000
Guilt	Violator	1.963	1.689	1.035	1.533	1.337	0.770	0.889
	Strong Tie	1.960	1.279	0.000	0.544	1.273	0.773	0.961
	Weak Tie	1.955	–	–	–	0.918	0.000	0.000
	Stranger	1.955	–	–	–	0.000	0.000	0.000
Embarrassment	Violator	1.952	1.186	0.865	1.555	1.150	0.989	0.622
	Strong Tie	1.971	1.116	1.130	1.469	1.091	1.110	0.783
	Weak Tie	1.920	1.095	0.639	1.132	1.194	1.120	0.956
	Stranger	1.829	0.896	0.990	1.305	1.319	1.460	0.934
Contempt	Violator	1.865	1.215	1.092	1.415	1.362	0.972	0.913
	Strong Tie	1.973	1.385	1.241	1.687	1.327	0.947	0.918
	Weak Tie	1.958	1.200	1.235	1.384	1.354	0.661	0.592
	Stranger	1.930	1.450	1.393	1.506	1.460	0.879	0.469
Disgust	Violator	1.915	1.021	0.934	1.499	1.178	1.264	1.370
	Strong Tie	1.915	1.737	1.446	1.612	1.425	1.123	1.450
	Weak Tie	1.985	1.524	1.488	1.626	1.495	1.108	1.127
	Stranger	1.986	1.731	1.615	1.747	1.647	1.131	1.138
Anger	Violator	1.887	1.289	0.972	1.408	1.509	0.790	1.095
	Strong Tie	1.952	1.539	1.379	1.488	1.384	0.895	1.023
	Weak Tie	1.978	1.392	1.368	1.482	1.464	0.802	0.979
	Stranger	1.980	1.606	1.246	1.547	1.628	1.141	0.919
Pride	Violator	1.945	1.059	0.610	1.268	1.275	0.522	0.764
	Strong Tie	1.946	1.128	0.750	0.684	1.325	0.000	1.299
	Weak Tie	1.924	0.650	0.000	–	1.459	0.391	0.000
	Stranger	1.762	0.000	0.000	–	0.000	0.439	0.000
Compassion	Violator	1.931	1.317	0.755	1.549	0.918	1.066	0.995
	Strong Tie	1.940	1.543	1.228	1.488	1.533	0.935	1.152
	Weak Tie	1.754	1.336	1.123	1.545	1.661	1.095	0.763
	Stranger	1.524	1.378	1.133	1.783	1.449	1.157	1.078

Table A17: Entropy of emotion intensities across roles and models. Missing values are shown as “–”. Shannon entropy (H) is computed over intensity values 1-4 only, capturing how varied the intensity is once an emotion is selected. The zero category was excluded from the entropy calculation, as it reflects whether the emotion was chosen at all rather than its intensity, and tends to dominate the distribution. Humans consistently show the highest entropy (~1.9-2) across nearly all emotions and roles, meaning when humans select an emotion, they spread intensities fairly evenly across 1-4. This holds regardless of emotion type or social distance. All LLMs show lower entropy than humans (except for Gemini-Compassion-Stranger case: 1.783 vs 1.524), confirming they concentrate on fewer intensity levels. The gap is substantial: most LLMs fall between 0.0-1.7, compared to humans’ near-uniform ~1.9. GPT-5.2 and Gemini tend to have the highest entropy among LLMs (~1.2-1.7), making them closest to human variability, while still falling short.

A.7.5 Individual Emotion Evaluation

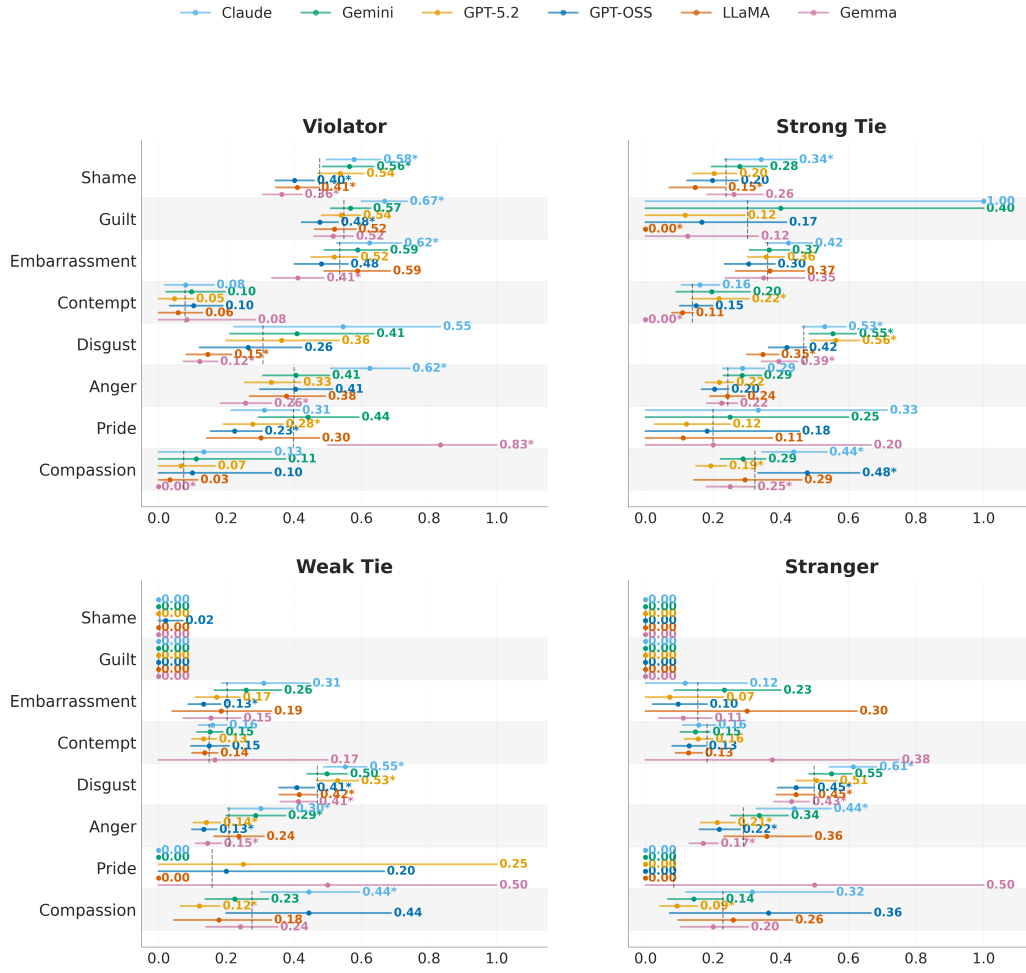


Figure A7: Models' precision scores for predicting emotional responses across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

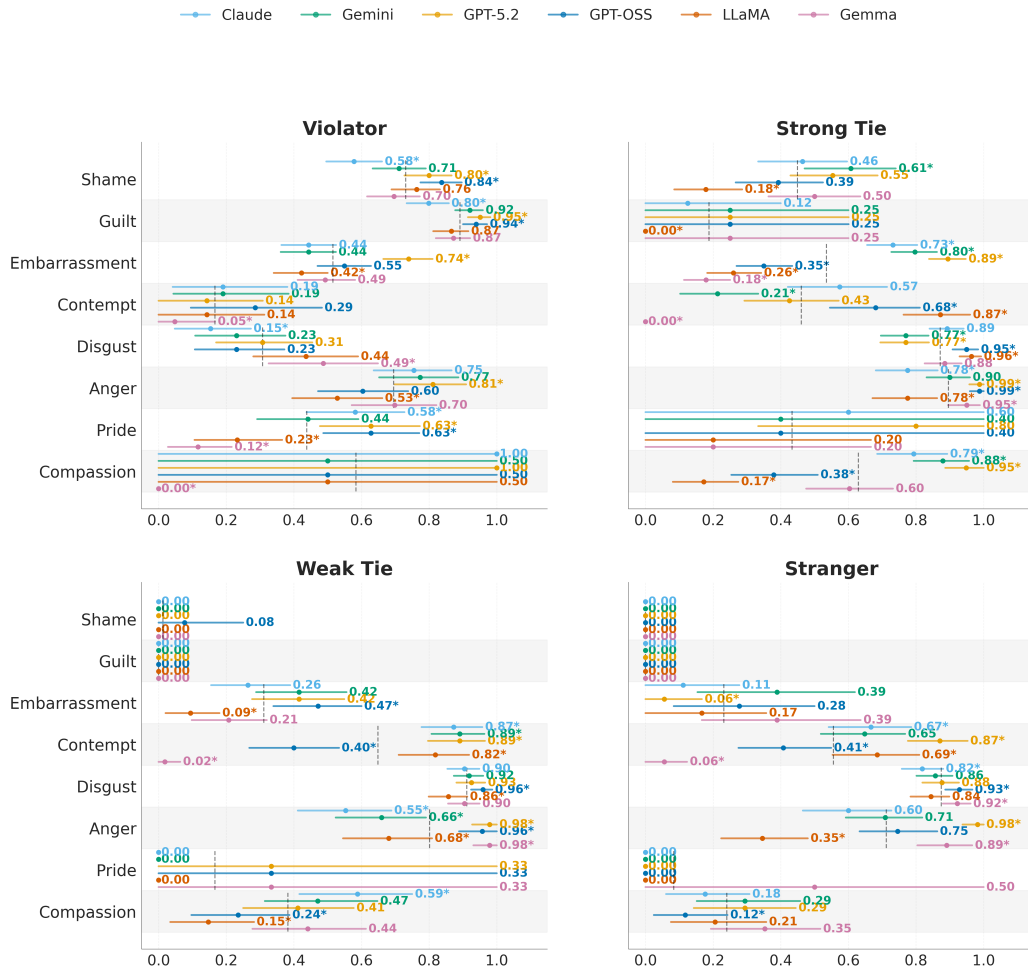


Figure A8: Models' recall scores for predicting emotional responses across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

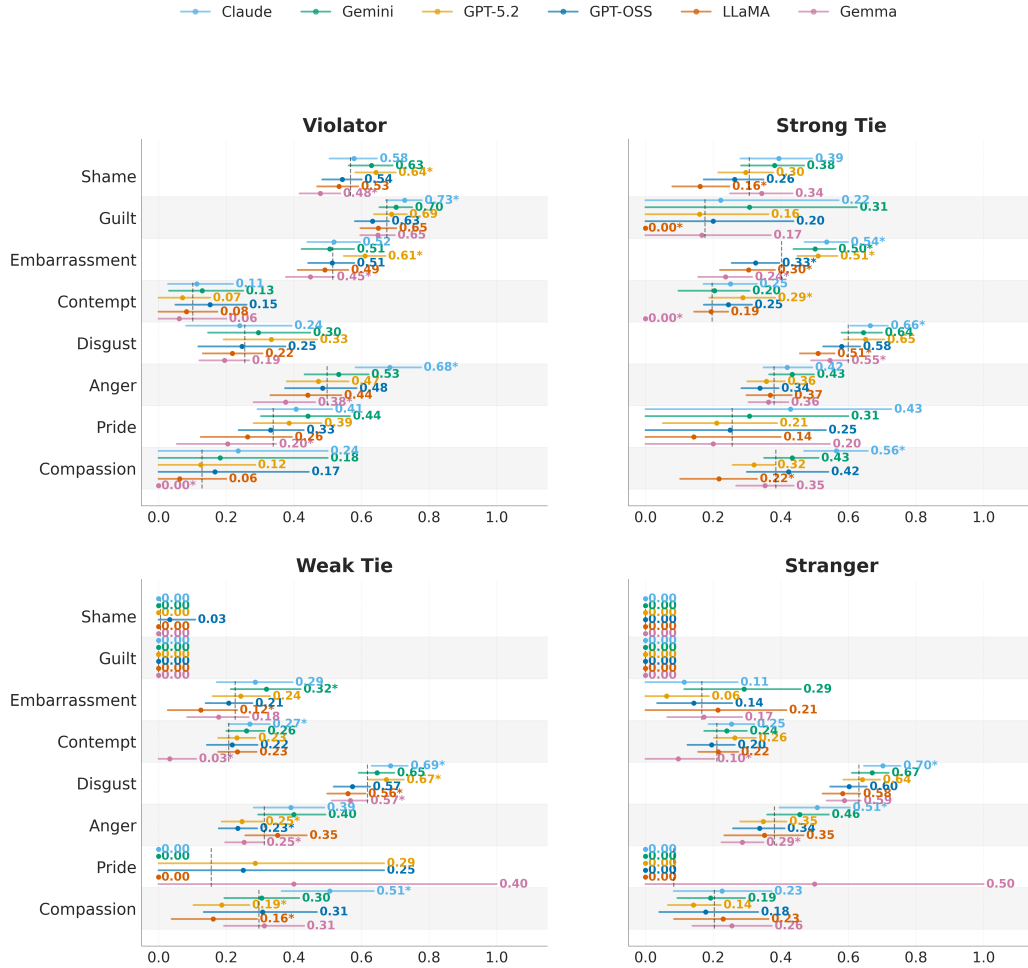


Figure A9: Models' F1 scores for predicting emotional responses across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

A.7.6 Individual Action Evaluation



Figure A10: Models' precision scores for predicting behavioral responses (would and should) across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

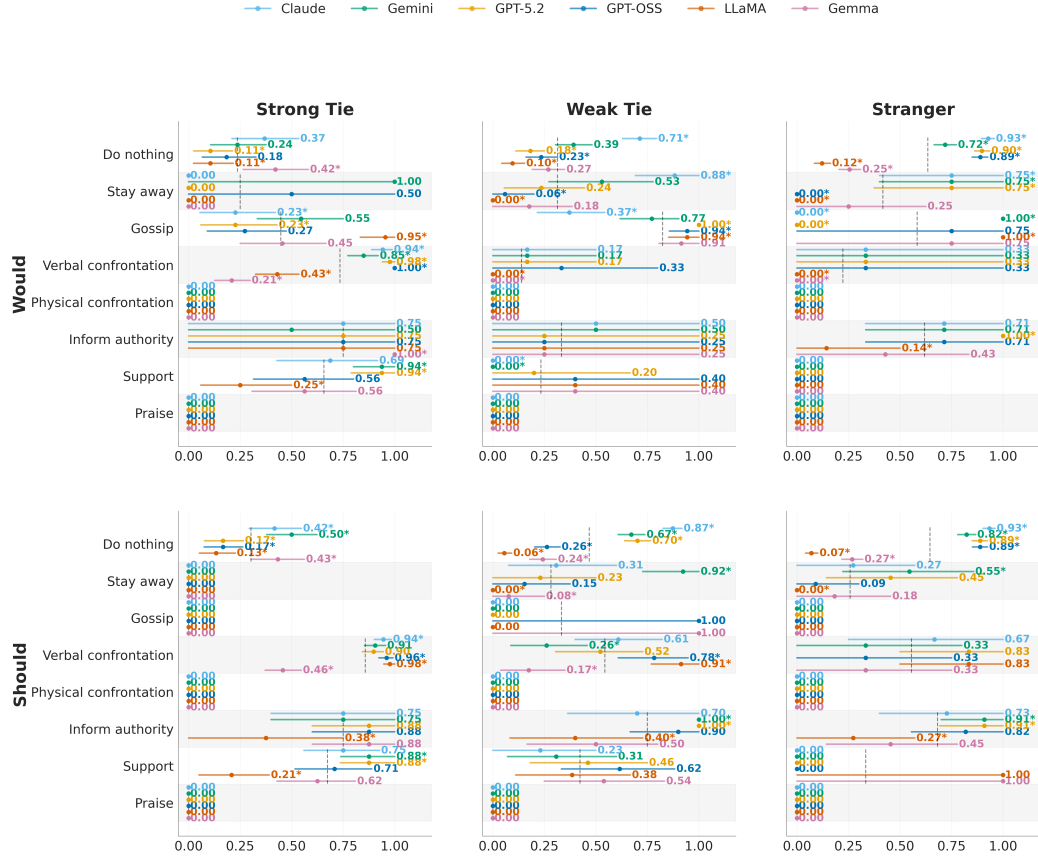


Figure A11: Models' recall scores for predicting behavioral responses (would and should) across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

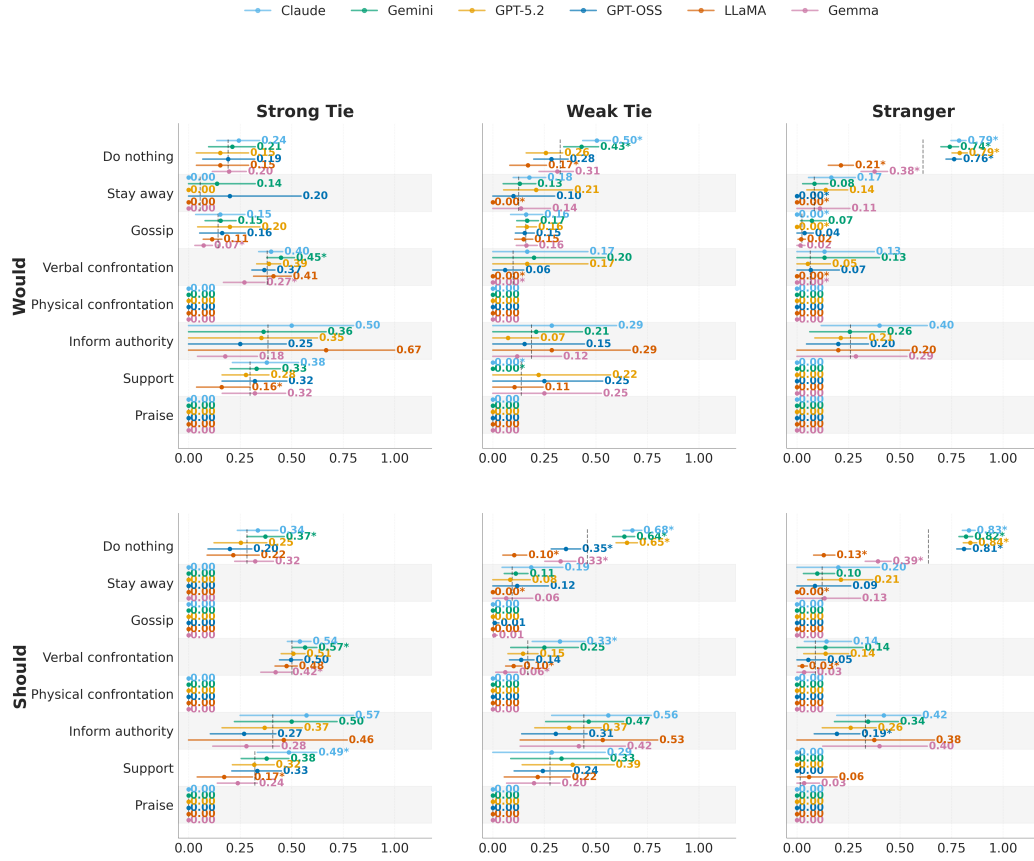


Figure A12: Models' F1 scores for predicting behavioral responses (would and should) across social relationships respectively. Dashed lines indicate mean performance across models. Points and error bars represent model estimates and 95% bootstrap confidence intervals relative to these means. * indicates the model differs significantly from the group mean ($p < 0.05$).

A.8 Supplemental Analysis

Deciding between classification and pairwise alignment To quantify the **alignment between emotional profiles** of two agents for the same stimulus, we computed the Euclidean distance between their emotion vectors for each stimulus across all possible emotions.

$$\text{Euclidean distance} = \sqrt{\sum_{k=1}^K (x_k - y_k)^2}$$

where x and y are the vectors of emotion strength ratings for the two agents across K emotions. Smaller Euclidean distances indicate greater similarity between emotional profiles, whereas larger values indicate greater dissimilarity. Each pair of agents was classified according to the identity of the rater, including: 1) human–human, where both responses were from human participants; 2) LLM–human, where one response was from a human and the other from a specific LLM (Claude, GPT-5.2, Gemini, GPT-OSS, Gemma, or LLaMA), with results reported separately for each model; and 3) LLM–LLM, where both responses were from LLMs, with pairwise combinations distinguished for each model pairing (e.g., Claude–GPT-5.2, Gemini–LLaMA, etc.). To ensure order-invariance (e.g., human–GPT-5.2 and GPT-5.2–human are treated identically), agent names were alphabetically sorted within each pair. Table A18 depicts the average Euclidean distance by pair type and by specific LLM and LLM pairing, highlighting differences between human–human pairs, each LLM–human pairing, and each LLM–LLM pairing.

Comparison	Actor	Strong Tie	Weak Tie	Stranger
Claude – Human	0.498	0.461	0.476	0.493
Gemini – Human	0.474	0.454	0.466	0.484
Gemma – Human	0.453	0.431	0.460	0.482
GPT-5.2 – Human	0.488	0.408	0.433	0.463
GPT-OSS – Human	0.472	0.429	0.448	0.485
LLaMA – Human	0.464	0.418	0.452	0.505
Human – Human	0.459	0.443	0.463	0.491

Table A18: Mean scaled Euclidean distance between human and model emotion presence vectors for *norm violator* (self) and *observer* (other) perspectives, along with a composite overall similarity score computed as their average. Standard errors are uniformly small (approximately 0.01 across models). Higher values indicate stronger alignment with human judgments

We examined Euclidean distance as a function of pair type using a linear mixed-effects model. The analysis included only human–human pairs and model–human pairs, with the latter defined separately for each LLM (Claude, GPT-5.2, Gemini, GPT-OSS, Gemma, and LLaMA). The model was specified as follows:

$$\text{Euclidean distance}_{ij} = \beta_0 + \beta_1(\text{pair type}_{ij}) + u_{0j} + \epsilon_{ij},$$

where u_{0j} represents a random intercept for each stimulus j , accounting for repeated measurements within the same stimulus, and ϵ_{ij} is the residual error. The reference level for pair type was human–human. Models were estimated using restricted maximum likelihood (REML) in R (lme4 package).

Table A19 summarizes the linear mixed-effects models predicting similarity in emotional responses (Euclidean distance) across social conditions. Human–human pairs serving as the reference category. Negative coefficients indicating greater similarity (smaller distance) relative to human pairs and positive coefficients indicating lower similarity.

Results varied by social distance and rating target. For **actor**, Claude–human ($b = 0.04, p < .001$) and GPT-5.2–human ($b = 0.03, p = .001$) showed significantly greater distances than human pairs, indicating reduced alignment, while other models did not differ significantly. In **strong tie**, GPT-5.2–human ($b = -0.03, p < .001$) and LLaMA–human ($b = -0.02, p = .002$) exhibited significantly smaller distances than the human baseline, indicating greater alignment, whereas Claude–human showed a small but significant increase in distance ($b = 0.02, p = .010$). A similar pattern emerged for **weak tie**, where GPT-5.2–human again showed significantly greater similarity ($b = -0.03,$

Relationship	Predictor	Estimate	95% CI	<i>p</i>
Actor	Claude–human	0.04	0.02 – 0.06	< .001
	Gemini–human	0.02	-0.00 – 0.03	.082
	Gemma–human	-0.01	-0.02 – 0.01	.525
	GPT-5.2–human	0.03	0.01 – 0.05	.001
	GPT-OSS–human	0.01	-0.00 – 0.03	.125
	LLaMA–human	0.01	-0.01 – 0.02	.524
Strong Tie	Claude–human	0.02	0.00 – 0.03	.010
	Gemini–human	0.01	-0.00 – 0.03	.114
	Gemma–human	-0.01	-0.03 – 0.00	.161
	GPT-5.2–human	-0.03	-0.05 – -0.02	< .001
	GPT-OSS–human	-0.01	-0.03 – 0.00	.100
	LLaMA–human	-0.02	-0.04 – -0.01	.002
Weak Tie	Claude–human	0.02	0.00 – 0.03	.037
	Gemini–human	0.01	-0.01 – 0.02	.437
	Gemma–human	0.00	-0.01 – 0.02	.950
	GPT-5.2–human	-0.03	-0.04 – -0.01	< .001
	GPT-OSS–human	-0.01	-0.03 – 0.00	.133
	LLaMA–human	-0.01	-0.02 – 0.01	.314
Stranger	Claude–human	0.00	-0.01 – 0.02	.679
	Gemini–human	-0.00	-0.02 – 0.01	.569
	Gemma–human	-0.01	-0.02 – 0.01	.371
	GPT-5.2–human	-0.03	-0.04 – -0.01	.002
	GPT-OSS–human	-0.00	-0.02 – 0.01	.663
	LLaMA–human	0.02	-0.00 – 0.03	.061

Table A19: Linear mixed-effects models predicting similarity (Euclidean distance) across social conditions. Human–human pairs serve as the reference category. Coefficients represent differences between each LLM–human pair and the human baseline within each condition.

$p < .001$), while Claude–human showed a small increase in distance ($b = 0.02$, $p = .037$); other models did not differ significantly. In **stranger** contexts, GPT-5.2–human continued to show significantly smaller distances than human pairs ($b = -0.03$, $p = .002$), whereas no other models showed reliable differences from the baseline.

Overall, these results indicate that emotional alignment between LLMs and humans is **model- and context-dependent**. GPT-5.2 consistently shows greater similarity to human emotional responses across multiple social distance, whereas other models tend to show either no reliable difference or modest deviations from the human baseline depending on context.

We examined **alignment in behavioral choices** across human and LLM agents for each social distance (strong ties, weak ties and strangers) and type of social expectation (“descriptive” vs. “injunctive”). Because responses were categorical rather than vector-valued, similarity between two agents was operationalized as action agreement rather than a distance-based measure. For each pair of agents and each stimulus, action similarity was coded as 1 if both agents selected the same action and 0 otherwise. This measure captures whether two agents converged on the same behavioral choice, with higher values indicating greater alignment in action selection.

As with the emotion analyses, all agent pairs were classified as human–human, specific LLMs–human, or LLM–LLM, and pair order was treated as invariant. Action similarity scores were analyzed using mixed-effects models with random intercepts for stimulus to account for repeated comparisons within the same scenario. Agreement between two agents was quantified as the proportion of trials in which both agents selected the same action. Across conditions, human–human agreement ranged from 25.8% to 50.0%. In contrast, LLM–human agreement varied more widely across models and contexts, ranging from 9.3% to 64.9% (Table A20). To assess differences from the human baseline, we fit logistic mixed-effects models with a random intercept for each stimulus, including pair type as a

Condition	Pair Type	Agreement (%)	Estimate	SE	<i>z</i>	<i>p</i>	Odds Ratio
Stranger — Descriptive	Human–Human	42.1					
	Claude–Human	59.3	0.9999	0.166	6.03	< .001	2.718
	Gemini–Human	46.0	0.2139	0.162	1.32	.188	1.239
	Gemma–Human	19.6	-1.5245	0.182	-8.40	< .001	0.218
	GPT-5.2–Human	54.0	0.6813	0.163	4.17	< .001	1.976
	GPT-OSS–Human	48.2	0.3434	0.162	2.12	.034	1.410
	Llama–Human	11.8	-2.2747	0.204	-11.15	< .001	0.103
Stranger — Injunctive	Human–Human	50.0					
	Claude–Human	64.9	0.9334	0.172	5.42	< .001	2.543
	Gemini–Human	56.7	0.3981	0.167	2.38	.017	1.489
	Gemma–Human	16.0	-2.4094	0.197	-12.21	< .001	0.090
	GPT-5.2–Human	58.0	0.4824	0.168	2.87	.004	1.620
	GPT-OSS–Human	52.7	0.1488	0.166	0.89	.372	1.160
	Llama–Human	9.3	-3.1742	0.225	-14.09	< .001	0.042
Weak tie — Descriptive	Human–Human	30.1					
	Claude–Human	40.9	0.7164	0.175	4.10	< .001	2.047
	Gemini–Human	31.6	0.1004	0.177	0.57	.571	1.106
	Gemma–Human	28.7	-0.1037	0.179	-0.58	.563	0.902
	GPT-5.2–Human	29.6	-0.0399	0.179	-0.22	.823	0.961
	GPT-OSS–Human	29.1	-0.0717	0.179	-0.40	.689	0.931
	Llama–Human	25.1	-0.3689	0.182	-2.02	.043	0.692
Weak tie — Injunctive	Human–Human	40.5					
	Claude–Human	46.2	0.2902	0.152	1.91	.057	1.337
	Gemini–Human	42.0	0.0736	0.153	0.48	.630	1.076
	Gemma–Human	17.3	-1.4342	0.175	-8.20	< .001	0.238
	GPT-5.2–Human	38.7	-0.1006	0.154	-0.66	.513	0.904
	GPT-OSS–Human	22.0	-1.0854	0.167	-6.52	< .001	0.338
	Llama–Human	17.3	-1.4342	0.175	-8.20	< .001	0.238
Strong tie — Descriptive	Human–Human	25.8					
	Claude–Human	36.0	0.6553	0.167	3.93	< .001	1.926
	Gemini–Human	33.1	0.4804	0.168	2.86	.004	1.617
	Gemma–Human	25.3	-0.0273	0.173	-0.16	.874	0.973
	GPT-5.2–Human	36.0	0.6553	0.167	3.93	< .001	1.926
	GPT-OSS–Human	35.3	0.6154	0.167	3.68	< .001	1.850
	Llama–Human	25.3	-0.0273	0.173	-0.16	.874	0.973
Strong tie — Injunctive	Human–Human	33.3					
	Claude–Human	42.9	0.7227	0.176	4.11	< .001	2.060
	Gemini–Human	41.8	0.6414	0.176	3.64	< .001	1.899
	Gemma–Human	30.9	-0.1888	0.180	-1.05	.295	0.828
	GPT-5.2–Human	38.4	0.3950	0.177	2.24	.025	1.484
	GPT-OSS–Human	36.9	0.2782	0.177	1.57	.116	1.321
	Llama–Human	36.4	0.2446	0.177	1.38	.167	1.277

Table A20: **Behavioral Choice Agreement and Model–Human Alignment Across Social Distances and Norm Type.** Agreement (%) reflects the observed proportion of identical choices within each pair type. Human–human agreement serves as the baseline level of consensus, while each model–human agreement is compared against this baseline using logistic regression. Reported estimates are log-odds differences relative to human–human agreement within each condition; negative values indicate lower agreement than the human baseline, whereas positive values indicate higher agreement. Standard errors (SE), *z*-statistics, and *p*-values test whether each model–human pair differs significantly from the human baseline. Odds ratios provide an interpretable effect size, with values below 1 indicating reduced alignment and values above 1 indicating increased alignment relative to the human baseline.

fixed effect with human–human agreement as the reference category. Rather than aggregating across LLMs, each model was entered separately, allowing us to estimate model-specific deviations from human consensus.

Results revealed substantial heterogeneity across models. Some LLMs (e.g., Claude, GPT-5.2) showed consistently higher agreement with humans than the human baseline in several conditions, particularly for descriptive judgments, whereas others (e.g., Gemma, Llama) exhibited markedly lower agreement, especially for stranger and injunctive contexts. Models such as Gemini and GPT-OSS tended to fall closer to the human baseline, with smaller and often nonsignificant differences depending on the condition.

Across social distances, alignment patterns varied systematically. For descriptive judgments (what agents would do), several models showed equal or higher agreement than human–human pairs for weak ties and strong tie, but more mixed or reduced alignment for strangers. For injunctive judgments (what agents should do), divergence from the human baseline was more pronounced for some models, particularly in stranger and weak ties conditions, though others maintained comparable or higher agreement.

These findings indicate that behavioral alignment with humans is not uniform across LLMs but instead depends both on the specific model, the social distance and type of social expectations elicited. While many models approximate human behavioral expectations in descriptive settings, greater variability emerges in prescriptive judgments, where some models diverge substantially from human consensus. Agreement rates and model-specific effects are summarized in Table A20. Together, these results suggest while LLMs consistently predict actual human behavioral reactions, they show a marked divergence from human judgment when they prescribe actions compared to human judgments.

Classification vs Alignment. We finally formulate social reasoning as a classification problem rather than evaluating alignment solely through pairwise similarity between human and model responses. In preliminary analyses above, similarity-based metrics yielded uniformly high human–model alignment even when models differed substantially from humans in absolute response levels and in condition-specific descriptive patterns. This occurs because similarity measures primarily reward shared response structure or rank ordering, while being comparatively insensitive to systematic shifts in magnitude, thresholding, and prevalence. In our setting, these differences are theoretically consequential: norm reasoning depends not only on whether responses vary in the same direction, but on whether a violation activates self-regulation, other-regulation, or sanction at all. We therefore use classification tasks that directly test these socially meaningful distinctions and make model failures more interpretable across roles and social-distance conditions.

Temperature Sensitivity Analysis To assess the robustness of model behavior to sampling variability, we conducted a temperature sensitivity analysis using Claude-4.5-Opus. This choice was guided by practical constraints: the Gemini-3-Pro model used in our main experiments was deprecated at the time of this analysis, and GPT-5.2 does not expose temperature as a tunable parameter when reasoning is enabled. For open-source models, running controlled temperature experiments proved computationally impractical in our setup (Google Colab), as inference was slow and long-running jobs were frequently interrupted by runtime disconnects.

We computed the Shannon entropy of emotion intensity distributions across emotion–role pairs and compared model outputs to human responses at three temperature settings ($T \in \{0, 0.5, 1\}$). Human responses exhibited substantially higher entropy than model outputs across all conditions (mean entropy: Human = 1.911; $T=0$: 1.033; $T=0.5$: 1.032; $T=1$: 1.045), indicating greater diversity in human emotional expression.

To quantify divergence, we calculated the mean absolute difference $|H_{\text{model}} - H_{\text{human}}|$ across $N = 27$ emotion–role pairs:

- $T = 0$: 0.916 ± 0.081
- $T = 0.5$: 0.878 ± 0.070
- $T = 1$: 0.941 ± 0.086

All model configurations showed significantly lower entropy than humans (Wilcoxon signed-rank tests, $p < .001$ for all three comparisons), confirming consistent under-dispersion in model outputs.

However, differences across temperature settings were not statistically significant (Wilcoxon signed-rank tests: $T=0$ vs. $T=1$, $p = .420$; $T=0$ vs. $T=0.5$, $p = .766$; $T=0.5$ vs. $T=1$, $p = .083$). Although $T=0.5$ yields the lowest mean divergence from human entropy, the lack of significant differences suggests that increasing temperature does not meaningfully improve alignment with

human-like variability. These results indicate that the entropy gap between humans and models is driven by structural limitations in model behavior rather than sampling stochasticity.

Emotion	Level	Human	Temp = 0	Temp = 0.5	Temp = 1
Shame	Violator	1.947	0.959	1.023	1.022
	Strong Tie	1.953	1.262	1.228	1.230
	Weak Tie	1.967	–	–	0.000
	Stranger	1.875	–	–	–
Guilt	Violator	1.963	1.035	1.021	1.082
	Strong Tie	1.960	0.000	–	0.000
	Weak Tie	1.955	–	–	–
	Stranger	1.955	–	–	–
Embarrassment	Violator	1.952	0.865	0.893	0.875
	Strong Tie	1.971	1.130	1.126	1.161
	Weak Tie	1.920	0.639	0.839	0.576
	Stranger	1.829	0.990	0.987	0.975
Contempt	Violator	1.865	1.092	0.999	1.066
	Strong Tie	1.973	1.241	1.190	1.225
	Weak Tie	1.958	1.235	1.225	1.185
	Stranger	1.930	1.393	1.407	1.418
Disgust	Violator	1.915	0.934	0.918	0.964
	Strong Tie	1.915	1.446	1.479	1.460
	Weak Tie	1.985	1.488	1.453	1.445
	Stranger	1.986	1.615	1.593	1.563
Anger	Violator	1.887	0.972	0.986	0.981
	Strong Tie	1.952	1.379	1.370	1.373
	Weak Tie	1.978	1.368	1.270	1.395
	Stranger	1.980	1.246	1.192	1.265
Pride	Violator	1.945	0.610	0.654	0.744
	Strong Tie	1.946	0.750	0.722	0.619
	Weak Tie	1.924	0.000	0.000	0.000
	Stranger	1.762	0.000	0.000	0.000
Compassion	Violator	1.931	0.755	0.904	1.067
	Strong Tie	1.940	1.228	1.150	1.154
	Weak Tie	1.754	1.123	1.057	1.110
	Stranger	1.524	1.133	1.191	1.262

Table A21: Entropy of emotion intensities across roles for humans and Claude under different temperature settings. Missing values are shown as “–”.